

**STD – X**  
**Subject : Science**



CHAPTER 1

**Chemical Reactions  
and Equations**





# CHAPTER 1

# Chemical Reactions and Equations

Part – 7

Types of Chemical Reactions

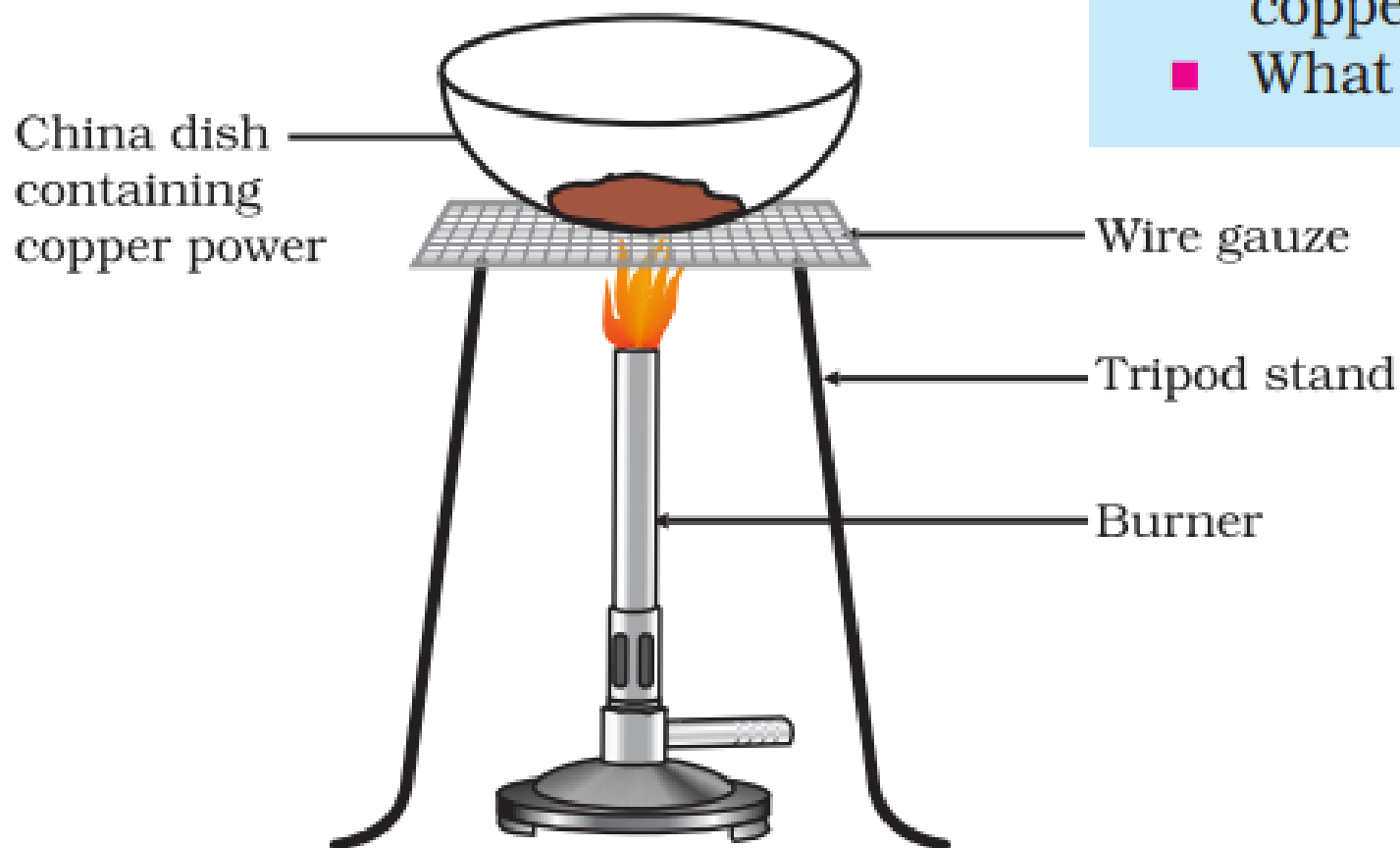
# **E) Oxidation - Reduction Reaction**



# Activity 1.11 – Oxidation and Reduction Reaction

## Activity 1.11

- Heat a china dish containing about 1 g copper powder (Fig. 1.10).
- What do you observe?



# Oxidation and Reduction

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graph TD; A[Oxidation and Reduction] --> B[Oxidation]; A --> C[Reduction];
```

## **Oxidation**

A chemical reaction in which  
a substance gains oxygen  
(or loses Hydrogen)  
is called oxidation reaction.

## **Reduction**

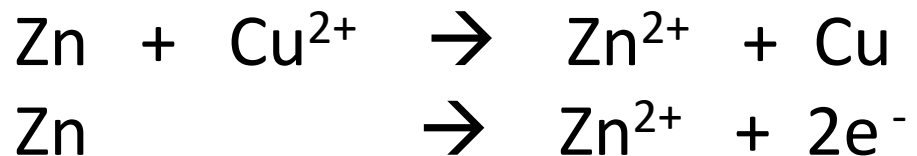
A chemical reaction in which  
a substance loses oxygen  
(or gains hydrogen)  
is called reduction reaction.

# Oxidation and Reduction

## Oxidation

Oxidation is defined as :

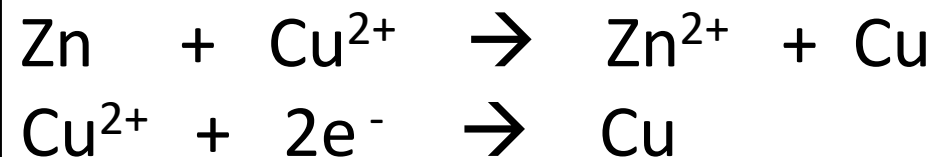
- gain of oxygen or
  - loss of hydrogen or
  - loss of electrons
- by a substance.



## Reduction

Reduction is defined as :

- gain of hydrogen or
  - loss of oxygen or
  - gain of electrons
- by a substance.



# E) Oxidation and Reduction

## Oxidation

Oxidation is defined as

- gain of oxygen or
  - loss of hydrogen or
  - loss of electrons
- by a substance.

## Reduction

Reduction is defined as

- gain of hydrogen or
  - loss of oxygen or
  - gain of electrons
- by a substance.

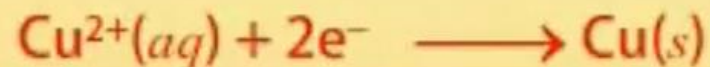
Zinc gets oxidised to  $\text{Zn}^{2+}$ , when a zinc rod is dipped in a copper sulphate solution.



In this reaction, zinc has lost two electrons to become  $\text{Zn}^{2+}$ . Thus, zinc gets oxidised.



On the other hand,  $\text{Cu}^{2+}$  gets reduced as it accepts two electrons (released by Zn) to become Cu.



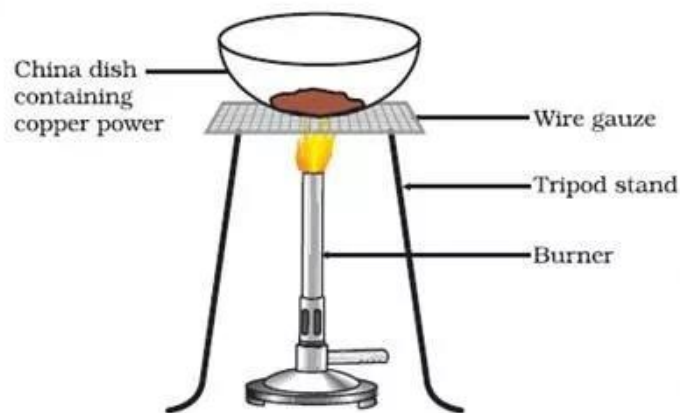
# E) Oxidation and Reduction

## OXIDATION : (Definition)

A chemical reaction in which a substance gains oxygen (or loses Hydrogen) is called oxidation reaction.

### Activity – 1.11 : Oxidation & Reduction

When brown coloured copper powder is heated in a china dish, then its surface gets coated with black copper (II) oxide.



In this reaction, oxygen is added to copper and copper oxide is formed. This is oxidation reaction.

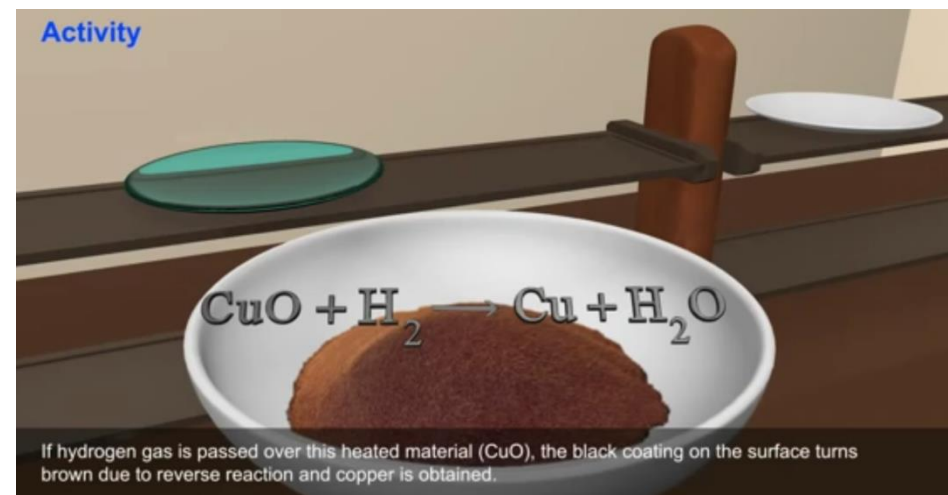
# E) Oxidation and Reduction

## REDUCTION : (Definition)

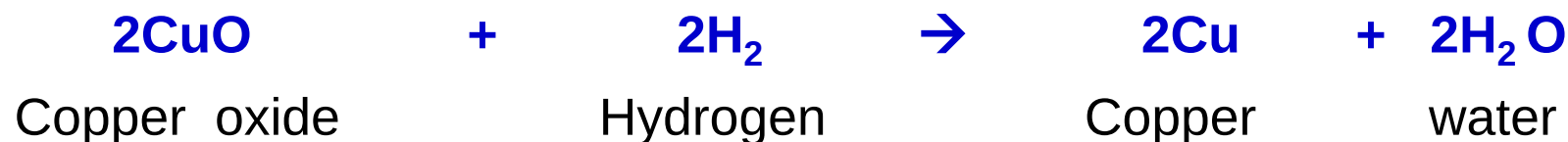
A chemical reaction in which a substance loses oxygen (or gains hydrogen) is called reduction reaction.

For example :

If hydrogen gas is passed over heated copper oxide (CuO) then black coating on the surface turns brown because oxygen is lost.



Heat



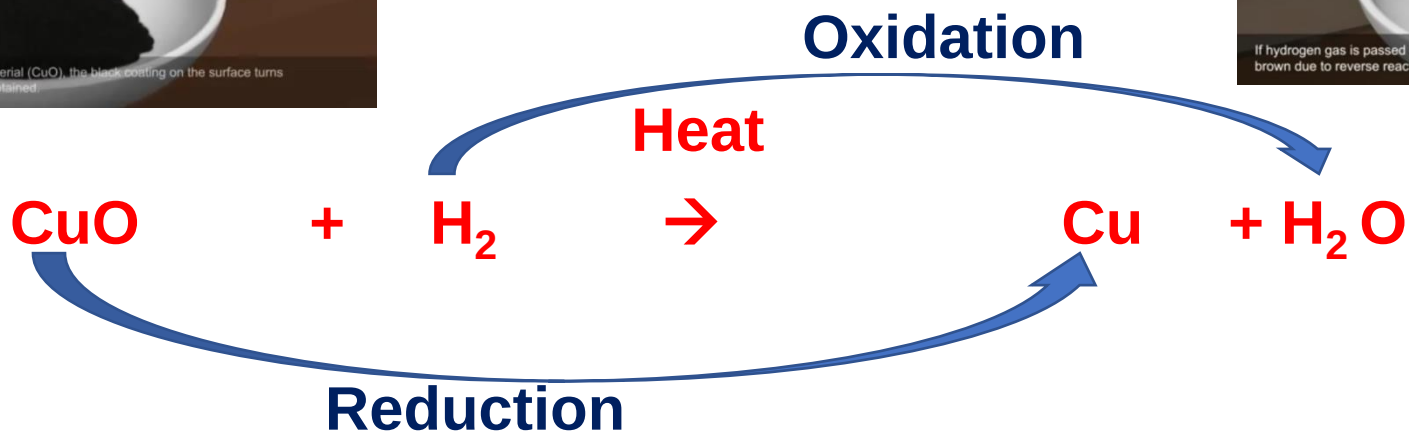
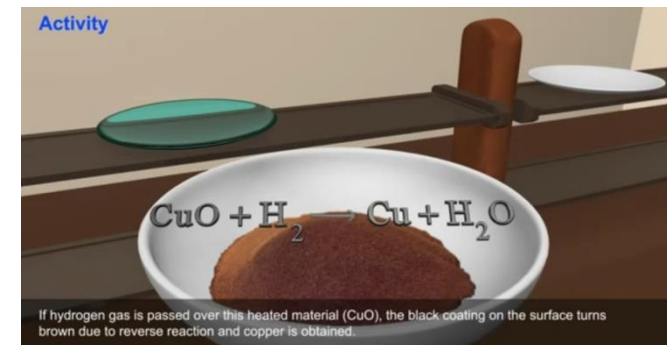
In this reaction, copper (II) oxide is losing oxygen and is being reduced. This is reduction reaction.

# E) Oxidation and Reduction

## OXIDATION – REDUCTION (Redox) REACTION : (Definition)

- A chemical reaction in which one reactant gets oxidised and the other reactant gets reduced is called oxidation-reduction reaction.
- Oxidation –Reduction reaction is also called as 'Redox – Reaction'.

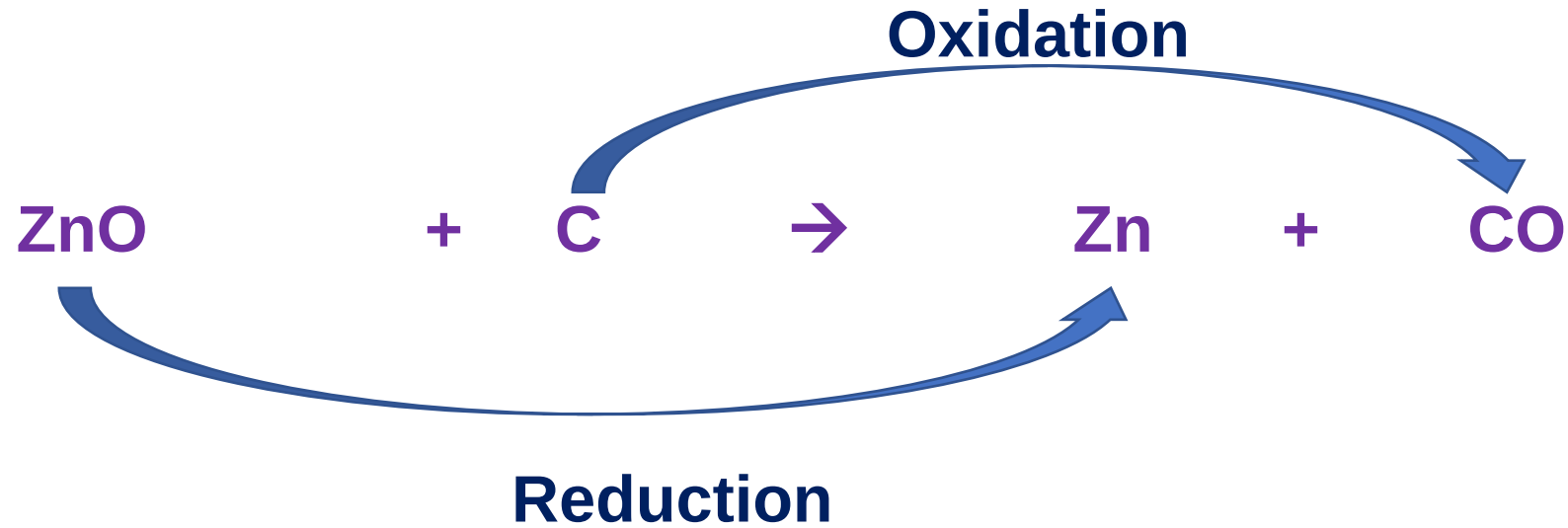
For example :



# Oxidation and reduction

Some other examples of Redox reaction :

When Zinc oxide reacts with carbon it forms zinc and carbon monoxide.

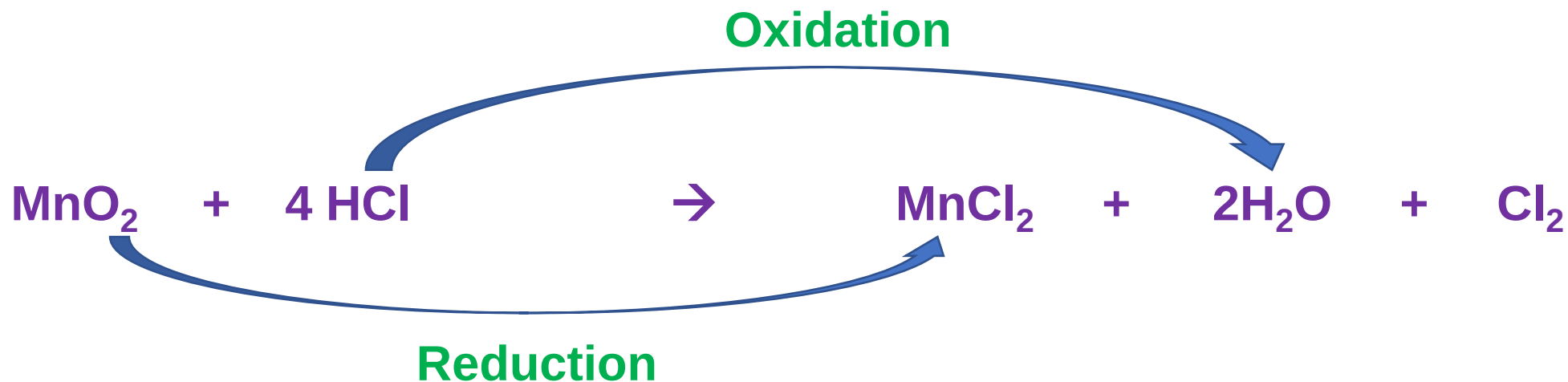


- Here,
- Carbon gets oxidised to form carbon monoxide.
- Zinc oxide gets reduced to form zinc.
- Therefore it is a Redox reaction.

# Oxidation and Reduction

Some other examples of Redox reaction :

- When manganese dioxide reacts with hydrochloric acid, it forms manganese chloride and chlorine gas



Here,

Hydrochloric acid gets **oxidised** to form water.

Manganese dioxide gets **reduced** to form manganese chloride.

Therefore it is a Redox reaction.

# Oxidation and Reduction

What did we learn?

- If a substance gains oxygen or loses hydrogen during a chemical reaction, it is called .....

Oxidation reaction

- If a substance loses oxygen or gains hydrogen during a chemical reaction, it is called .....

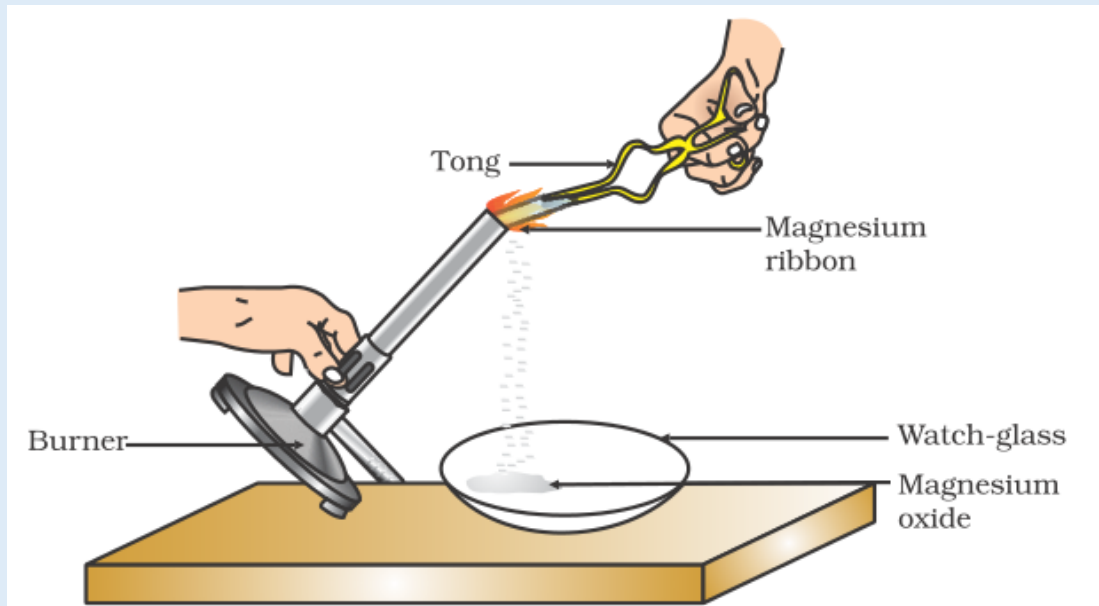
Reduction reaction

- A chemical reaction in which one reactant gets oxidised and the other reactant gets reduced is called .....

Oxidation - Reduction reaction or REDOX reaction

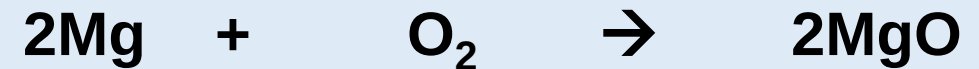
# Oxidation and Reduction

When magnesium ribbon burns with a dazzling flame in air (oxygen) and changes into a white substance –magnesium oxide.



**Figure 1.1**

*Burning of a magnesium ribbon in air and collection of magnesium oxide in a watch-glass*



• Is magnesium being oxidised or reduced in this reaction?

Ans : Magnesium gets oxidised in this reaction.

**TO BE CONTINUED .....**

